

Paper Title: Forecasting X with Y — A Comparative Evaluation

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Abstract

Depositvolume is a the amount of cash sitting inside the money market account. For a bank it is of great interest to predict the depositvolume, in order to know the amount of money they can work with. In this project we focused on different variables and machine learning methods to predict the depositvolume.

Placeholder abstract text (target: ≤ 150 words, no citations).

Contents

Since non-maturing deposits have no contractual maturity it is essential for banks to have a model predicting the volume. If they do not have such a model or the model does not work it is impossible for the liquidity management to work properly. After a brief introduction this paper will go over the data used to predict the deposit-volume NEEDS VARIABLE V_t .

1 Introduction

Research Question(s):

- RQ1: ...

2 Related Work

Table 1: Overview of related work and positioning of this paper.

Reference	Data/Market	Method	Rolling	Out-of-sample	...
Author et al. (Year)	...	...	✗	✓	...
Author et al. (Year)	...	...	✓	✗	...
This paper	...	...	✓	✓	...

3 Methods

3.1 Problem Setting and Data

All datapoints are based on January 2003 - December 2025:

- The overnight deposit volume and interest rate was sourced from Deutsche Bundesbank and consist of monthly end-of-period levels(?).
- The Euro Short-Term Rate €STR sourced from the ECB, consisting of data for each business day(?).
- The German geopolitical risk index GPRC_DEU sourced monthly(?).
- The month-on-month inflation MoM – Inflation sourced monthly(?).
- The DAX index sourced with yfinance on business days(?).
- The 10-year German bond yield 10Y Bond sourced on business days(?).

Overnight deposits or non-maturity deposits are deposits with next-day maturity. Consisting mainly of fully transferable deposits and deposits non-transferable deposits that can be converted by end of the next business day (?).

The target variable is the daily volume of overnight deposits V_t held by private household, sourced from Deutsche Bundesbank. Predicting the actual volume comes with a hand full of problems since:

- reaches a new high/low many times -> can't be predicted by many(/any??) machine learning algorithm - Solution: Transform the target variable into the difference in change between two days To get all of the data points into daily rates different methods have been applied to ensure the best possible data while reaching the wanted granularity:

ADD METHODS HERE SEE 5.2 Masterthesis

3.2 Exploratory Data Analysis

HERE PLOTS OF THE DATA ARE SHOWN

3.3 Models

$$\hat{y}_{t+h} = f(X_t, X_{t-1}, \dots, X_{t-p}; \theta) + \varepsilon_t \quad (1)$$

3.4 Evaluation Strategy

Figure 1: Overview of the analytics pipeline: data preprocessing, model training, and evaluation stages.

4 Results

Table 2: Out-of-sample performance comparison across models. Metrics computed on the reconstructed volume level and the differenced series (mean $\pm$ std over rolling origins).

Model	MAE	RMSE	R^2
Naive (Random Walk)	—	—	—
Naive (Hold)	—	—	—
Linear Regression	—	—	—
XGBoost	—	—	—
FFN	—	—	—
TFT	—	—	—

Figure 2: Example caption style: “Comparison of forecasting accuracy in terms of RMSE, measured from [start date] to [end date] across [list of models].”

5 Discussion

6 Conclusion and Outlook

References

- Caldara, D. and Iacoviello, M. (2022). Geopolitical risk index (*gprc_{deu}*). *Accessed* : [9.7.2026].
- Destatis (2025). Consumer price index (*cpi*) and inflation rates in germany. *Accessed*: [9.7.2026].
- Deutsche Bundesbank (2025a). Bank statistics: Deposits of domestic non-financial corporations and households. *Accessed*: [9.7.2026].
- Deutsche Bundesbank (2025b). Yields of debt securities outstanding: 10-year maturity. *Accessed*: [9.7.2026].
- Deutsche Börse (2025). Dax historical data. *Accessed*: [9.7.2026].

European Central Bank (2025). Euro short-term rate (€str). Accessed: [9.7.2026].

European Central Bank (2026). Overnight deposits. Monetary Policy Glossary, <https://www.ecb.europa.eu/services/glossary/html/act6o.en.html>. Accessed: [9.7.2026].

A Appendix: Additional Figures and Tables

B Appendix: Code

Listing 1: Example: model fitting excerpt

```
1 def fit_sequence_model(model_name, X_train, y_train, seed=SEED):  
2     # ...  
3     pass
```